

YUKTA BATRA

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TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, SQL, HTML, CSS
Frameworks & Libraries: Spring Boot, Node.js, React, Angular, Vue, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Keras
ML & AI: PyTorch, Deep Learning, Neural Networks, Natural Language Processing, Computer Vision, Data Pipelines
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, DynamoDB, ECS), Docker, Kubernetes, Jenkins, GitHub Actions, Azure, Linux, Git
Concepts: REST APIs, Microservices Architecture, Distributed Systems, System Design, Object-Oriented Programming (OOP), Data Structures & Algorithms, Agile, CI/CD, Unit Testing, Serverless
Certifications: AWS Cloud Practitioner, Linux Foundation OpenSSF (LFEL1012) – Secure AI/ML Software Development, Machine Learning, Data Visualization (Tableau), JavaScript & ReactJS Bootcamp

EXPERIENCE

Graduate Teaching Assistant – George Mason University, Fairfax, VA Aug 2025 – May 2026

- Engineered reusable functions and automated data pipelines in R to migrate 50+ legacy STATA scripts, cutting processing time by 60% and establishing reproducible analytical workflows.
- Streamlined analysis workflows across 5+ datasets in R, generating reproducible visualizations and structured reports consumed by 30+ graduate students per semester.
- Performed code review and debugging on 30+ R scripts per cycle, identifying logic errors and refactoring for modularity; reduced error rates by 40% and improved script performance.

Open Source Contributor – MITRE Secure Coding Case Studies Mar 2026 – Apr 2026

- Proposed and authored a secure coding case study on CVE-2025-55315 (HTTP Request Smuggling in ASP.NET Core Kestrel) for MITRE’s public repository, analyzing the parser vulnerability and mitigation via PR #64037.

Web Developer Intern – MakeMeBuilders, India Jun 2022 – Aug 2022

- Delivered responsive React applications serving client-facing interfaces, reducing page load time by 30% through streamlined component rendering and lazy loading.
- Developed modular UI components, accelerating feature development by 20% across all project iterations.
- Converted 15+ Figma designs into production-ready components, reducing design-to-code handoff time by 35% and accelerating stakeholder approval cycles.

Cloud Computing Intern – AWS Academy Cloud Foundations, India Jun 2021 – Jul 2021

- Provisioned and configured multi-tier AWS infrastructure (EC2, S3, RDS, IAM) across 5+ hands-on lab projects, deploying scalable cloud application environments from scratch.
- Executed AWS Well-Architected Framework principles across 5 pillars — security, reliability, performance efficiency, cost optimization, and operational excellence — in project-based cloud deployments.
- Configured auto-scaling, application load balancing, and VPC networking for high-availability deployments; earned AWS Cloud Practitioner certification.

PROJECTS

AI Resume Screener | *Python, FastAPI, Groq LLaMA 3.3, pypdf, Vercel, Render*

- Built a full-stack AI-powered resume screening tool using Groq’s LLaMA 3.3 to analyze resume-job description match, extract skill gaps, detect missing ATS keywords, and generate actionable improvements.
- Deployed FastAPI backend on Render and static frontend on Vercel; processes PDF parsing and LLM inference in under 5 seconds.

Cloud-Native Survey Management Platform | *Spring Boot, React, Docker, Kubernetes, AWS (EC2, RDS, S3), Jenkins*

- Architected a full-stack survey system supporting 1,000+ concurrent submissions using stateless Spring Boot microservices and RESTful APIs.
- Orchestrated Docker deployments on Kubernetes with zero-downtime rolling updates; configured Amazon RDS with strategic indexing achieving 30% faster query performance.
- Automated CI/CD pipelines using Jenkins and GitHub Actions, reducing deployment cycles by 40%.

GMU SmartPatriot – AI-Powered University Knowledge Engine | *Next.js, Node.js, TypeScript, Groq (Llama-3.1), Cheerio, Vercel*

- Deployed an AI knowledge engine aggregating 200+ GMU web pages via a RAG pipeline combining web scraping, semantic retrieval, and natural language processing.
- Integrated multi-turn conversation memory (5–7 turns) into a serverless full-stack interface on Vercel, maintaining context continuity with <2-second response latency.

Human Activity Recognition (HAR) using CNN–LSTM | *Python, PyTorch, Pandas, NumPy*

- Constructed an end-to-end data pipeline for time-series classification using smartphone IMU data across 14 activity classes.
- Designed a hybrid CNN–LSTM deep learning model achieving 98% test accuracy and 0.97 macro F1 on ~6,000 windows.
- Validated real-time inference on unseen live sensor recordings, achieving consistent class predictions within sub-second execution time.

Additional Projects: Network Address Manager (C/AVL Trees), Simulated Annealing, Maze Solver (A* Search) — [Github](#)

EDUCATION

Master of Science in Computer Science Aug 2024 – May 2026
George Mason University, Fairfax, VA
Relevant Courses: Algorithms, Artificial Intelligence, Mathematics, Information Security

Bachelor of Technology in Computer Science Aug 2019 – Jun 2023
The NorthCap University, Gurugram, India
Relevant Courses: Object-Oriented Design, Database Systems, Operating Systems